

EXPERIMENT 25: FRUIT AND COLOR USING BACKTRACKING

Aim

To write a **Prolog program to identify fruits and their colors using backtracking.**

Algorithm

1. Start.
2. Define fruit and color facts.
3. Associate each fruit with its color.
4. Query the database for a fruit or color.
5. Prolog uses backtracking to find all matching solutions.
6. Display the results.
7. Stop.

Source Code

% Fruit and Color Database

fruit_color(apple, red).

fruit_color(banana, yellow).

fruit_color(orange, orange).

fruit_color(grapes, green).

fruit_color(mango, yellow).

fruit_color(guava, green).

Queries and Output

?- fruit_color(Fruit, yellow).

Fruit = banana ;

Fruit = mango.

?- fruit_color(mango, Color).

Color = yellow.

Output

```
?- [fruit_color].  
% fruit_color.pl compiled 0.00 sec, 6 clauses  
  
?- fruit_color(Fruit, yellow).  
Fruit = banana ;  
Fruit = mango.  
true.  
  
?- fruit_color(mango, Color).  
Color = yellow.  
true.  
  
?- 
```

Result

The Prolog program successfully identifies fruits and their colors using **backtracking** to retrieve multiple matching solutions.